

A Low-Cost Prototype Classroom Attendance Checker and Logger Using Facial Recognition

Ian Gabriel C. Santillan

*Ateneo de Davao University Senior High School, Davao City, PH
igcsantillan@addu.edu.ph*

ABSTRACT: Facial recognition is a classification of biometric software that maps a person's facial highlights scientifically and stores it as a faceprint; they work by contrasting chosen facial highlights from a given picture with faces inside a database. Commercially available systems are costly and do not provide the full user control of the system. This study aimed to make a Facial Recognition and Data Logging system using the Internet of Things to create a flexible system to allow users maximum control of the system. The device has undergone Functionality and Accuracy Tests, which tested the Setup Time, Recognizing Time, and Accuracy of the device. Results showed that the researcher successfully created a facial recognition device with a 100% functionality rating and an accuracy rating of 87% with an average setup time of 4.339 seconds and an average recognition time of 5.790 seconds. The results imply that the device can be compared to commercially available facial recognition systems in terms of basic functionalities while allowing maximum ability to modify the program due to the open-source nature of the Raspberry Pi. Maintaining it at a lower price by 33.4% can potentially increase the system's availability to be used by more people for security or data logging purposes.

KEYWORDS: Facial Recognition, Database, Data Logging, Internet of Things, Raspberry Pi

Introduction. In recent years, manual checking of professor's presence or attendance in the classroom in their respective class hours has been causing disturbances due to opening of doors and it is prone to personal errors. This is also the same with checking student's attendance in the classroom. It is also very unproductive for the checkers to roam around the campus, checking every classroom from time to time and for the teachers checking their student's attendance. Also, the effectiveness of the present system may decrease due to the newly employed checkers not being familiar with some of the professors. A false report or hearsay might also lead to wrong statement and this is considered as a factor that can be attributed as confusion to the class itself. There arises a need for a more efficient and effective method of checking the punctuality and attendance of the professors and students (Cruz, et al. 2015).

A facial recognition system is a technology equipped for recognizing or confirming an individual from a digital image or a video frame from a video source. There are different techniques in which facial recognition systems work, however in general, they work by looking at chosen facial features from given picture with face prints inside a database. It is additionally depicted as a Biometric Artificial Intelligence based application that can exceptionally distinguish an individual by analyzing patterns dependent on the individual's facial features and

shape (Heinzman 2019) (Symanovich n.d.) (Techopedia 2020).

High cost, limited flexibility as well as limited accessibility are some of the problems that are faced by current commercially available biometric systems which include the facial recognition systems and the products associated with the function (Thakkar 2018).

The device, can recognize facial prints and log the data in a Comma Delimited file which can easily be accessed by teachers and school officials for checking of class attendance. It is controlled using peripherals connected to the Raspberry Pi where you can also see live video feed from the device's camera. It also features a motion tracking function to track object movement during the facial recognition process. Since commercially available biometric systems are costly and has limited flexibility for development (Thakkar 2018), the device can potentially be a substitute to commercially available systems while maintaining a low cost. The device is capable of taking facial prints, training the machine learning model with those features, and recognizing those features while simultaneously logging the data.

Background Information. Facial recognition is a classification of biometric software that maps a person's facial highlights scientifically and stores the information

as a face print. The product utilizes deep learning algorithms to think about a live catch or computerized picture then puts away the face print with the end goal of confirming a person's identity (Rouse 2018). Facial recognition systems vary widely in price and functionality. Commercially available systems found online would range around \$140-\$200 (Indiamart n.d.).

Deep learning also known as deep structured learning or hierarchical learning is part of a broader family of machine learning methods based on learning data representations, as opposed to task-specific algorithms. Learning can be supervised, semi-supervised or unsupervised (Schmidhuber 2015) (Bengio, Courville and Vincent 2013) (Bengio, LeCun and Hinton 2015).

One key favorable position of a facial recognition framework is that it can provide mass identification as it does not require the cooperation of the test subject to work. Appropriately planned frameworks introduced in airplane terminals, multiplexes, and other open spots can recognize people among the group, without passers-by being aware of the system ("Top Five Biometrics: Face, Fingerprint, Iris, Palm and Voice" 2017). Applications for the system include advanced human-computer interaction, video surveillance, automatic indexing of images, and video database, among others (Bramer 2006).

Engineering Goals. The device must be able to be carry out basic facial recognition through data sets provided by the researcher. The device must also be able to store or log the data in a spreadsheet program such as Excel that is easily accessible. The device should be able to integrate its motion tracking function through its sensors. The device should also be programmed in an open-source ecosystem for maximum flexibility and control of the device. The total cost of the finished device will be kept under \$100 so that it remains cost-effective and can be implemented in public schools.

Methods. The research methods – data collection via facial recognition, data organization, statistical analysis, software development, and electronic design construction - were undertaken using the following software packages: OpenCV (OpenCV n.d.), JASP (JASP n.d.), Arduino IDE (Arduino n.d.), Python IDLE 3 (Python n.d.) (OpenCV n.d.), and Microsoft Office Excel (Microsoft n.d.).

Mechanics.

The body of the device (Figure 1) utilized a 6.3'' x 3.74'' x 2.17'' phone box as the compartment for the electronic components of the device. The components were arranged with the ports facing the sides of the box to

allow additional hardware such as USB cable, HDMI cable, and power cables to be connected. Holes were cut on the box to provide access to these ports and three additional holes were cut on the top of the box for the jumper cables, Raspberry Pi camera ribbon, and for the output spline of the servo motor.

The sensors and the camera were held by a modified ultrasonic sensor holder using hot glue and two 1'' x 0.5'' scrap prototype boards. The two ultrasonic sensors were attached on both ends with hot glue while maintaining an angle of 130°. The servo horn was then attached to the bottom of the ultrasonic sensor holder using hot glue.

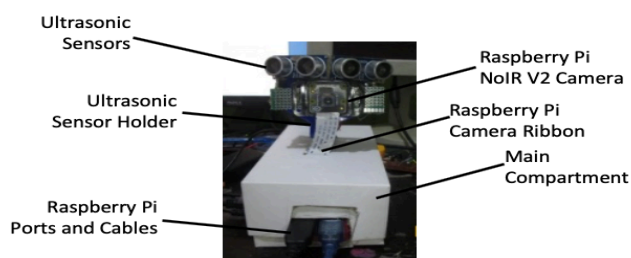


Figure 1. Device Front View

Electronics.

All the electronics were attached inside the compartment (Figure 2 and 3) of the device interior with hot glue, and the wires were secured and organized with zip ties. The signal wires of the ultrasonic sensors and servo were connected to header pins soldered onto the prototype board. Jumper wires were also soldered onto the prototype board and directed to the respective signal pins on the Arduino microcontroller. The Raspberry Pi NoIR V2 camera was connected directly to the camera module port. The Arduino microcontroller was powered via a USB cable connected to the Raspberry Pi. The Raspberry Pi was then powered using a 5V, 2.5 Raspberry Pi B+ Power Supply.



Figure 2. Arduino Components



Figure 3. Raspberry Pi Components

The device was controlled and monitored using peripherals connected to the Raspberry Pi which includes: a keyboard, a mouse, and a monitor. Data recognized by the device was then stored in a Comma Delimited file which can be accessed using Excel.

Camera and Sensors.

The Raspberry Pi Camera was mounted on the modified holder along with the ultrasonic sensors. The camera provides live feed to the monitor as well as live dataset construction and live facial recognition. The dataset construction was initiated by running the dataset script where the camera captures pictures for the dataset this can also be done manually by uploading pictures directly to the dataset folder. The training script is then run before executing the facial recognition script which compares values stored as faceprints in the datasets to labels, in this case names.

The device also has two ultrasonic sensors which compares the distance of the object from each sensor. The data is then sent to the Arduino which then send instructions for servo motor to rotate in the direction closer to one of the sensors. This was ultimately for the motion tracking function of the device.

Code.

The code to control the servo motor and the ultrasonic sensors for the motion tracking function of the device written in Arduino. Much of the code for the function was (Arduino n.d.)adapted from open sources such as Instructibles (Kielas-Jensen 2016). The facial recognition function for the camera was written in the Python 3 IDLE within the Raspberry Pi OS while also utilizing OpenCV 3 Library. The code for the machine learning program was adapted from Jacky Le's Raspberry-Facial-Recognition scripts (Le 2017), which was modified to show user labels

and to log data directly to a Comma Delimited file. The scripts were mainly for each machine learning function namely live dataset construction, dataset training, and facial recognition. The device functioned by executing these Python scripts on the Raspberry Pi.

Testing.

The device was tested for three trials per repetition for three repetitions each to test the functionality of the servo motor and the camera. The functionality of the ultrasonic sensors was also tested for three repetitions, each repetition consisted of object distances 5 cm, 10 cm, 15 cm, 20 cm, and 25 cm. The servo motor and the camera were functional all throughout the tests, while the ultrasonic sensors were functional until the 25 cm mark.

The facial recognition system was tested using three tests namely: the setup time test, the recognition time test, and the accuracy test. The datasets were divided into two groups, the first one will be used for training while the second one will be used for testing. The datasets consisted of five unique face print to be utilized for the training and the testing. The results for the setup time and recognition time were measured in seconds, while the accuracy test was enumerated. The data was recorded and then went through basic statistical analysis using the JASP software.

Results. Fifteen setup times and fifteen recognition times were collected in total from the setup and recognition tests. The accuracy of the device was also tested for five unique face prints in three different configurations for a total of fifteen recorded results.

As shown in Figure 4, the bar graph shows the setup times of the facial recognition system for all five unique face prints and their three different configurations. The mean value was 4.37s while the minimum and maximum values were, 2.44s and 8.04s respectively.

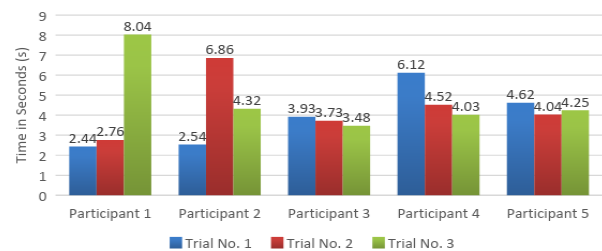


Figure 4. Setup Time Test

Figure 5 shows the bar for the recognition times of the facial recognition system for all five unique face prints and their three different configurations. The mean value was 4.33s while the minimum and maximum values were, 2.18s and 7.46s respectively.

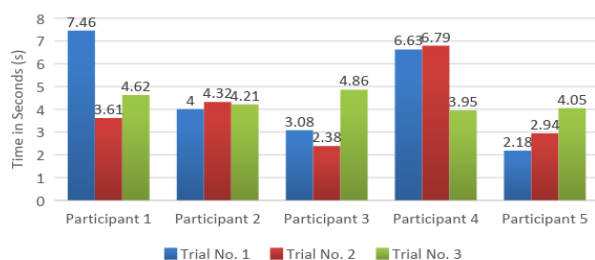


Figure 5. Recognition Time Test

Lastly, Figure 6 then shows the pie graph for the accuracy of the facial recognition system for all five unique face prints and their three different configurations. The system was able to recognize 13 out of the 15 unique face print configurations therefore achieving an accuracy of 87%. The results were checked by checking the Comma Delimited files while specific face prints were showed to know if they were registered or not.

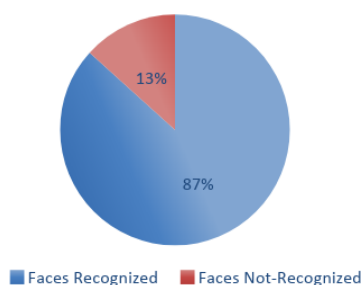


Figure 1. Accuracy Test

Discussion. Overall, the device met the engineering goals and performed effectively in both the functionality tests and the facial recognition system tests. The device was controlled via peripheral devices connected to the Raspberry Pi, and the live video stream allowed the user to check the accuracy of the device regarding its facial recognition function. The overall functionality was successful as the device was able to recognize unique facial prints while simultaneously executing the motion tracking function and the data logging function. The total cost of the robot was \$93.24, which is 6.76% under the 100-dollar budget and comparatively less expensive by 33.4% than commercially available facial recognition system which is priced at around \$140-\$200. It was successfully tested for its setup time and recognition time with the mean values being 4.37s and 4.33s. The accuracy of the facial recognition system was rated at 87%.

Going forward, some modifications can be made to improve the capabilities of the device. A better camera

can be used with better quality to increase feature detection. Furthermore, the cameras do not function well in low visibility environments. The visibility can be improved upon the addition of a light source. The facial recognition system can also be modified to increase the accuracy of the facial recognition system. A bigger dataset could have also been integrated to train the model better in recognizing facial prints. This project can be further developed with the addition of a Graphic User Interface (GUI) for easy setup and access of the device. expansion of the robot's autonomous capabilities. Currently, the device was tested in a controlled environment, but implementation in a classroom or school setting is possible. There is also potential for the addition of a GSM Module to allow real-time messaging from the device to the guardians of students.

Conclusion. While it certainly had limitations, the device met the budget requirement and was comparatively lesser than its commercial counterparts by around 33.7%. The device was able to do its programmed function without any errors or malfunctions found during the functionality tests. The device was also able to successfully identify, recognize facial features and log the data with 100% functionality and 87% accuracy with an average setup time of 4.378s and an average recognition time of 4.339 s. The device was also able to accomplish the goal of the research which is for the device to be programmed in an open-source ecosystem for the maximum flexibility and control of the system. There is potential for improvement regarding lighting, GUIs, and the facial recognition system itself, which will make the device more suitable for implementation in classroom or school settings. With its current limitations, the device performs most effectively in environments with high visibility and high numbers of training datasets. The device has promising applications in integrating a complex system such as facial recognition at a lower cost.

▪ Acknowledgements

The author thanks Henry Haranay, Romel Villarubia, Florentino Manuel Jr. of Tagum City National High School and Engineer Chris Malecdan of Saint Louis University for assisting as research mentors. The author also thanks his upperclassmen, Francine Racho Joseph Nino Boyles, Kurt Cabinian, and Jex Cansancio for their advice and guidance. To Sharon Mae Santillan and Rhynell Santillan for their support and for funding the research. To Nicole Rylen Astillo for her constant support. And to the mighty God who guided and protected

and guided the author to finish the research successfully and safely.

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▪ Author

Ian Santillan is a rising senior at Ateneo de Davao University Senior High School, Davao City, PH. His research interests include Machine Learning, Artificial Intelligence, Computational Biology, Ecological Modeling and Marine Biology.